



# Vaxcel®

## ACYCLOVIR 250mg

### I.V. FOR INFUSION

#### COMPOSITION:

Each vial contains  
Acyclovir (as Acyclovir Sodium) 250mg

#### PRESENTATION:

White to off white crystalline powder. The reconstituted solution is clear.

#### INDICATIONS:

Vaxcel Acyclovir 250mg I.V. for Infusion is indicated:

1. For the treatment of *Herpes simplex* infections.
2. For the prophylaxis of *Herpes simplex* infections in immune-compromised patients.
3. In the treatment of *Varicella zoster* infections.
4. For the treatment of *Herpes simplex* infections in the neonate.

#### PHARMACODYNAMICS:

##### Microbiology:

Acyclovir is an antiviral agent which is active in vitro against Herpes simples (HSV) types I & II and Varicella zoster virus (VZV). However, the relationship between in vitro sensitivity of herpes viruses to acyclovir and clinical response to therapy has yet to be established. Acyclovir needs to be phosphorylated to the active compound acyclovir triphosphate, in order to become active against the virus. Such conversion is very limited in normal cells and in addition, cellular DNA polymerase is not very sensitive to the active compound. However, in infected cells, HSV or VZV coded thymidine kinase facilitates the conversion of acyclovir to acyclovir monophosphate which is then converted to acyclovir triphosphate by cellular enzymes. Acyclovir triphosphate acts as an inhibitor of, and substrate for, the herpes specified DNA polymerase preventing further viral DNA synthesis. Animal studies indicate that at high doses acyclovir is cytotoxic.

#### PHARMACOKINETICS:

In adults, mean steady state peak plasma concentrations (C<sub>ssmax</sub>) following a one-hour infusion of 2.5mg/kg, 5mg/kg, 10mg/kg and 15 mg/kg were 22.7 micromoles (5.1 micrograms/ml), 43.6 micromoles (9.8 micrograms/ml), 92 micromoles (20.7 micrograms/ml) and 105 micromoles (23.6 micrograms/ml), respectively. The corresponding trough levels (C<sub>ssmin</sub>) 7h later

were 2.2 micromoles (0.5 micrograms/ml), 3.1 micromoles (0.7 micrograms/ml), 10.2 micromoles (2.3 micrograms/ml) and 8.8 micromoles (2.0 micrograms/ml), respectively. In children over 1 year of age similar mean peak (C<sub>ssmax</sub>) and trough (C<sub>ssmin</sub>) levels were observed when a dose of 250mg/m<sup>2</sup> was substituted for 5mg/kg and a dose of 500 mg/m<sup>2</sup> was substituted for 10mg/kg. In neonates (0 to 3 months of age) treated with doses of 10mg/kg administered by infusion over a one-hour period every 8h the C<sub>ssmax</sub> was found to be 61.2 micromoles (13.8 micrograms/ml) and the C<sub>ssmin</sub> to be 10.1 micromoles (2.3 micrograms/ml).

##### Distribution:

Cerebrospinal fluid levels are approximately 50% of corresponding plasma levels. Plasma protein binding is relatively low (9 to 33%) and drug interactions involving binding site displacement is not anticipated.

##### Elimination:

In adults the terminal plasma half life of acyclovir after administration of acyclovir i.v. for infusion is about 2.9h. Most of the drug is excreted unchanged by the kidney. Renal clearance of acyclovir is substantially greater than creatinine clearance indicating that tubular secretion in addition to glomerular filtration contributes to the renal elimination of the drug. 9-carboxymethoxy-methylguanine is the only significant metabolite of acyclovir and accounts for approximately 10 to 15% of the dose excreted in the urine. When acyclovir is given one hour after 1 gram of probenecid the terminal half life and the area under the plasma concentration time curve is extended by 18% and 40% respectively. In neonates (0 to 3 months of age) treated with doses of 10mg/kg administered by infusion over a one-hour period every 8h the terminal plasma half life was 3.8h.

##### Special Patient Populations:

In patients with chronic renal failure the mean terminal half life was found to be 19.5h. The mean acyclovir half life during haemodialysis was 5.7h. Plasma acyclovir levels dropped approximately 60% during dialysis. In the elderly total body clearance falls with increasing age, associated with decreases in creatinine clearance, although there is little change in the terminal plasma half life.

#### DOSAGE AND ADMINISTRATION:

A course of treatment with Acyclovir intravenous Infusion usually lasts 5 days, but this may be adjusted according to the patient's condition and response to therapy. Treatment for herpes encephalitis and neonatal *Herpes simplex* infections usually lasts 10 days. The duration of prophylactic administration of Acyclovir intravenous infusion is determined by the duration of the period at risk.

##### Dosage in adults:

###### Treatment of *Herpes simplex*

Obese patients should be dosed at the recommended adult dose using ideal body weight, rather than actual body weight. Patients with *Herpes simplex* (except herpes encephalitis) should be given Acyclovir intravenous infusion in doses of 5 mg/kg body weight every 8 hours. Patients with *Herpes encephalitis* should be given Acyclovir intravenous infusion in doses of 10mg/kg body weight every 8 hours provided renal function is not impaired.

##### Prophylaxis of *Herpes simplex* in immune-compromised patients

Obese patients should be dosed at the recommended adult dose using ideal body weight, rather than actual body weight. Refer to adult dosing recommendations for the treatment of *Herpes simplex* with Vaxcel Acyclovir 250mg I.V. for Infusion.

##### Treatment of *Varicella* and *Herpes zoster*

Obese patients should be dosed at the recommended adult dose using ideal body weight, rather than actual body weight. Patients with *Varicella zoster* infections should be given Acyclovir intravenous infusion in doses of 5mg/kg body weight every 8 hours. Immune-compromised patients with *Varicella zoster* infections should be given Acyclovir intravenous infusion in doses of 10mg/kg body weight every 8 hours provided renal function is not impaired (see Dosage in renal impairment).

##### Dosage in children:

The dose of Acyclovir Intravenous Infusion for children aged between 3 months and 12 years is calculated on the basis of body surface area. Children with *Herpes simplex* (except *Herpes encephalitis*) or *Varicella zoster* infections should be given Acyclovir intravenous infusion in doses of 250mg per square metre of body surface area every 8 hours. In immune-compromised children with *Varicella zoster* infections or children with *Herpes encephalitis*, Acyclovir intravenous infusion should be given in doses of 500mg per square meter body surface area every 8 hours if renal function is not impaired. Children with impaired renal function require an appropriately modified dose, according to the degree of impairment.

##### Dosage in neonates:

The dosage of Acyclovir intravenous infusion in neonates is calculated on the basis of body weight. Neonates with *Herpes simplex* infections should be given Acyclovir intravenous infusion in doses of 10mg/kg body weight every 8 hours.

##### Dosage in the elderly:

The possibility of renal impairment in the elderly must be considered and dosage should be adjusted accordingly (see *Dosage in renal impairment* below). Adequate hydration should be maintained.

##### Dosage in renal impairment:

Caution is advised when administering Acyclovir Intravenous Infusion to patients with impaired renal function. Adequate hydration should be maintained.

The following adjustments in dosage are suggested:

| Creatinine Clearance | Dosage                                                                                |
|----------------------|---------------------------------------------------------------------------------------|
| 25 to 50 ml/min      | The dose recommended above (5 or 10mg/kg body weight) should be given every 12 hours. |
| 10 to 25 ml/min      | The dose recommended above (5 or 10mg/kg body weight) should be given every 24 hours. |

0 (anuric) to 10 ml/min

In patients receiving continuous ambulatory peritoneal dialysis (CAPD) the dose recommended above (5 or 10mg/kg body weight or 500mg/m<sup>2</sup>) should be halved and administered every 24 hours and after dialysis.

In patients receiving haemodialysis the dose recommended above (5 or 10mg /kg body weight) should be halved and administered every 24 hours and after dialysis.

#### ROUTE OF ADMINISTRATION:

Vaxcel Acyclovir 250mg I.V. for Infusion contains no preservative. Should visible turbidity or crystallization appear in the solution before or during infusion, the preparation should be discarded. Reconstituted or diluted solutions SHOULD NOT BE REFRIGERATED.

##### Administration:

The required dose of Vaxcel Acyclovir 250mg I.V. for Infusion should be administered by slow intravenous infusion over one-hour period.

##### Reconstitution:

Since no antimicrobial preservative is included, reconstitution and dilution must be carried out under full aseptic conditions, immediately before use. Vaxcel Acyclovir 250mg I.V. for Infusion should be reconstituted using the 10ml volumes of either Water for Injections BP or Sodium Chloride Injection BP (0.9%w/v) to provide a solution containing 25mg/ml acyclovir.

For the calculated dose, determine the appropriate number and strength of vials to be used. To reconstitute each vial, add the recommended volume of infusion fluid and shake gently until the contents of the vial have dissolved completely. Alternatively, the reconstituted solution may be further diluted to give an acyclovir concentration of not greater than 5mg/ml (0.5%w/v) for administration by infusion:-

Add the required volume of reconstituted solution to the chosen infusion solution, as recommended below, and shake well to ensure adequate mixing occurs. For children and neonates, where it is advisable to keep the volume of fluid to a minimum, it is recommended that dilution is on the basis of 4ml reconstituted solution (100mg acyclovir) added to 20ml of infusion fluid.

For adults, it is recommended that infusion bags containing 100ml of infusion fluid are used, even when this would give an acyclovir concentration substantially below 0.5%w/v. Thus one 100ml infusion bag may be used for any dose between 250mg and 500mg acyclovir (10 and 20ml of reconstituted solution) but a second bag must be used for doses between 500mg and 1000mg.

When diluted in accordance with the recommended schedules, Vaxcel Acyclovir 250mg I.V. for Infusion is known to be compatible with the following infusion fluids and stable

for up to 48 hours below 25°C, beyond that unused solution should be discarded:-

- Water for Injection;
- Sodium Chloride Intravenous Infusion BP (0.9%w/v);
- Sodium Chloride (0.45% w/v) and Glucose (4%w/v)

Intravenous Infusion BP;

- Compound Sodium Lactate Intravenous Infusion BP (Hartmann's Solution)

Vaxcel Acyclovir 250mg I.V. for Infusion when diluted in accordance with the above schedule will give an acyclovir concentration not greater than 0.5%w/v.

#### CONTRAINDICATIONS:

Vaxcel Acyclovir 250mg I.V. for Infusion is contraindicated in patients known to be previously hypersensitive to acyclovir or valacyclovir.

#### WARNING & PRECAUTIONS:

In patients receiving Vaxcel Acyclovir 250mg I.V. for Infusion at higher doses (e.g. for *Herpes encephalitis*), specific care regarding renal function should be taken, particularly when patients are dehydrated or have any renal impairment. Reconstituted Vaxcel Acyclovir 250mg I.V. for Infusion has a pH of approximately 11.0 and should not be administered by mouth.

*Use in patients with renal impairment and in elderly patients:*

Acyclovir is eliminated by renal clearance; therefore the dose must be reduced in patients with renal impairment. Elderly patients are likely to have reduced renal function and therefore the need for dose reduction must be considered in this group of patients. Both elderly patients and patients with renal impairment are at increased risk of developing neurological side effects and should be closely monitored for evidence of these effects. In the reported cases, these reactions were generally reversible on discontinuation of treatment.

*Use in pregnancy:*

Acyclovir was found not to be teratogenic. Acyclovir can be excreted in breast milk. Hence Acyclovir should not be used in pregnancy or nursing mothers unless the potential benefit justifies the potential risk to the fetus and nursing infant.

#### INTERACTION WITH OTHER MEDICAMENTS:

Acyclovir is eliminated primarily unchanged in the urine via active renal tubular secretion. Any drugs administered concurrently that compete with this mechanism or affect renal physiology may increase acyclovir plasma concentrations. Probenecid and cimetidine increase the AUC of acyclovir by this mechanism, and reduce acyclovir renal clearance. However no dosage adjustment is necessary because of the wide therapeutic index of acyclovir. In patients receiving intravenous acyclovir, caution is required during concurrent administration with drugs which compete with acyclovir for elimination, because of the potential for increased plasma levels of one or both drugs or their metabolites. Increases in plasma AUCs of acyclovir and of the inactive metabolite of mycophenolate mofetil, an immunosuppressant agent used in transplant patients, have been shown when the

drugs are co-administered. Care is also required (with monitoring for changes in renal function) if administering intravenous acyclovir which affect other aspects of renal physiology (e.g. cyclosporine, tacrolimus). There are reports of additive nephrotoxicity when both acyclovir and cyclosporine are administered concomitantly.

*Incompatibilities:* None Known

#### SIDE EFFECTS:

| System                                          | Frequency | Adverse Events                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Blood and lymphatic system disorder             | Uncommon  | Decreases in haematological indices (anemia, thrombocytopenia, leukopenia)                                                                                                                                                                                                                                                   |
| Immune system disorders                         | Very rare | Anaphylaxis                                                                                                                                                                                                                                                                                                                  |
| Psychiatric and nervous system disorders        | Very rare | Headache, dizziness, agitation, confusion, tremor, ataxia, dysarthria, hallucinations, psychotic symptoms, convulsions, somnolence, encephalopathy, coma.                                                                                                                                                                    |
| Vascular disorders                              | Common    | Phlebitis                                                                                                                                                                                                                                                                                                                    |
| Respiratory, thoracic and mediastinal disorders | Very rare | Dyspnoea                                                                                                                                                                                                                                                                                                                     |
| Gastrointestinal disorders                      | Common    | Nausea, vomiting                                                                                                                                                                                                                                                                                                             |
|                                                 | Very rare | Diarrhoea, abdominal pain                                                                                                                                                                                                                                                                                                    |
| Hepato-biliary disorders                        | Common    | Reversible increases in liver-related enzymes                                                                                                                                                                                                                                                                                |
|                                                 | Very rare | Reversible increases in bilirubin, jaundice, hepatitis                                                                                                                                                                                                                                                                       |
| Skin and subcutaneous tissue disorders          | Common    | Pruritus, urticaria, rashes (including photosensitivity)                                                                                                                                                                                                                                                                     |
|                                                 | Very rare | Angioedema                                                                                                                                                                                                                                                                                                                   |
| Renal and urinary disorders                     | Common    | Increases in blood urea and creatinine. Rapid increases in blood urea and creatinine levels are believed to be related to the peak plasma levels and the state of hydration of the patient. To avoid this effect the drug should not be given as an intravenous bolus injection but by slow infusion over a one-hour period. |

|                                                      |           |                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | Very rare | Renal impairment, acute renal failure. Adequate hydration should be maintained. Renal impairment usually responds rapidly to rehydration of the patient and/or dosage reduction or withdrawal of the drug. Progression to acute renal failure, however can occur in exceptional cases. |
| General disorders and administration site conditions | Very rare | Fatigue, fever, local inflammatory reactions. Severe local inflammatory reactions sometimes leading to breakdown of the skin have occurred when Vaxcel Acyclovir 250mg I.V. for Infusion has been inadvertently infused into extracellular tissues.                                    |

#### SYMPTOMS AND TREATMENT OF OVERDOSE:

*Symptoms and Signs:*

Overdosage of intravenous acyclovir has resulted in elevations of serum creatinine, blood urea nitrogen and subsequent renal failure. Neurological effects including confusion, hallucinations, agitation, seizures and coma have been described in association with overdosage.

*Treatment:*

Patients should be observed closely for signs of toxicity. Haemodialysis significantly enhances the removal of acyclovir from the blood and may, therefore, be considered a management option in the event of symptomatic overdose.

#### STORAGE:

Store below 30°C.

After reconstitution, store below 25°C for 48 hours. Do not refrigerate. Protect from light.

Keep out of reach of children

Jauhi dari kanak-kanak

#### PACK QUANTITIES:

Available in one vial & 10 vials of sterile powder for intravenous infusion per box.



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